

Yuval Boker

Software Engineer | Backend & Systems | Python, C++, Linux

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Summary

Software Engineer specializing in backend and systems development using C++ and Python on Linux. Experience building RESTful APIs and database-driven systems, with a strong focus on performance, reliability, and debugging. Background in system design and low-level programming through selective excellence training. Particularly interested in systems internals, code analysis, networking fundamentals, and cybersecurity concepts, with a strong analytical and problem-solving mindset.

Education

B.Sc. in Computer Science, Tel-Hai University (2022–2025)

- Selected for Excellence team – a highly selective program for outstanding computer science students focused on system design, debugging, and advanced software development in Python and C++ on Linux.
- Coursework: Algorithms, Data Structures, Operating Systems, Software Engineering, Artificial Intelligence.

Professional training

Excellence team – Startup Nation Central

Selective excellence program focused on advanced software engineering. Designed and implemented end-to-end software solutions in Python and C++, with emphasis on system design, debugging, and maintainability. Presented technical projects to senior engineers from leading tech companies.

Projects

- **Linux File System – MyFS (C++):** Designed and implemented a user-space file system backed by a virtual block device stored in a regular file. Built layered architecture including block device abstraction, VFS layer, and file system logic with directory and metadata management, supporting core file operations on Linux.
- **Google Project – Python:** Designed and implemented an autocomplete engine using optimized data structures for efficient lookup and memory usage, emphasizing correctness across diverse input scenarios.
- **Basketball Shot Detector – Python:** Developed a real-time video processing system for basketball shot detection using motion and color-based techniques, API integration, and deployment support on Linux.

Technical Skills

- **Programming Languages:** Python · C++ · JavaScript · Java · C · TypeScript
- **Backend & APIs:** Node.js · RESTful APIs · HTTP/HTTPS · Client–Server Architecture · Microservices
- **Systems & Software Engineering:** Linux · Operating Systems · System Design · Debugging · Algorithms
- **Databases:** PostgreSQL · MongoDB · SQL Data Modeling
- **Tools & Technologies:** Docker · Git · OpenCV (cv2) · LLM APIs (OpenAI, Claude, Claude Code)

Experience

Teaching Assistant – Tel-Hai University (2025)

- Mentored students in algorithms and analytical coursework, reviewing solutions and providing structured feedback to strengthen problem-solving, debugging skills, and algorithmic thinking.

Military Service

Military Service – IDF, Magal Unit (2018–2021)

- Served as a commander, leading teams, training personnel, and coordinating operations in high-pressure environments.

Languages

- Hebrew – Native | English – Proficient.